Chapter 12. Generalized Least Squares

Correlated and heteroscedastic errors, GLS estimator, properties

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12.1 The general linear model

So far the errors were uncorrelated with equal variance. In time series, clustered or heteroscedastic data this fails. The **general linear regression model** is

$$\mathbf{y} = X\beta + \epsilon, \qquad \epsilon \sim N_n(\mathbf{0}, \sigma^2 V),$$

with V a known positive definite matrix, X of size $n \times (p+1)$ and full column rank. Ordinary least squares (OLS) is no longer the BLUE.

Since V is symmetric positive definite, $V = P\Lambda P'$ with P orthogonal, and $T = P\Lambda^{1/2}$ satisfies $V = TT'$. Multiply the model by T^{-1}:

$$\mathbf{z} = W\beta + \eta, \qquad \mathbf{z} = T^{-1}\mathbf{y}, \quad W = T^{-1}X, \quad \eta = T^{-1}\epsilon,$$

and $\operatorname{Var}\eta = \sigma^2 T^{-1}V(T^{-1})' = \sigma^2 T^{-1}TT'(T')^{-1} = \sigma^2 I_n$. The transformed model satisfies the usual assumptions.

Theorem 12.1 (Generalized least squares)

The BLUE of β is the **generalized least squares** (GLS) estimator

$$\hat\beta_G = (W'W)^{-1}W'\mathbf{z} = (X'V^{-1}X)^{-1}X'V^{-1}\mathbf{y},$$

which minimizes $(\mathbf{y}-X\beta)'V^{-1}(\mathbf{y}-X\beta)$. It satisfies $E\hat\beta_G = \beta$ and $\operatorname{Var}\hat\beta_G = \sigma^2(X'V^{-1}X)^{-1}$.

Proof. Apply the Gauss–Markov theorem (Theorem 7.1) to the transformed model; linear unbiased estimators in $\mathbf{z}$ are the same as those in $\mathbf{y}$, since T is invertible. Using $(T^{-1})'T^{-1} = (TT')^{-1} = V^{-1}$, $W'W = X'V^{-1}X$ and $W'\mathbf{z} = X'V^{-1}\mathbf{y}$. Also $\|\mathbf{z}-W\beta\|^2 = (\mathbf{y}-X\beta)'V^{-1}(\mathbf{y}-X\beta)$. Finally $E\hat\beta_G = (X'V^{-1}X)^{-1}X'V^{-1}X\beta = \beta$ and $\operatorname{Var}\hat\beta_G = (X'V^{-1}X)^{-1}X'V^{-1}(\sigma^2V)V^{-1}X(X'V^{-1}X)^{-1} = \sigma^2(X'V^{-1}X)^{-1}$. ∎

When V is diagonal, $V = \operatorname{diag}(v_i)$, GLS is **weighted least squares** with weights $w_i = 1/v_i$: minimize $\sum w_i(y_i - \mathbf{x}_i'\beta)^2$, giving less weight to noisier observations.

12.2 Properties

OLS under correlated errors. The OLS estimator $\hat\beta = (X'X)^{-1}X'\mathbf{y}$ is still unbiased, but

$$\operatorname{Var}\hat\beta = \sigma^2(X'X)^{-1}X'VX(X'X)^{-1},$$

which is at least $\sigma^2(X'V^{-1}X)^{-1}$ (by Gauss–Markov), with equality when $V = I$ (more generally, when the column space of X is invariant under V).

Residuals and σ^2. The GLS residual vector is $\mathbf{e}_G = \mathbf{y} - X\hat\beta_G = \{I - X(X'V^{-1}X)^{-1}X'V^{-1}\}\mathbf{y}$, with

$$E\mathbf{e}_G = \mathbf{0}, \qquad \operatorname{Var}\mathbf{e}_G = \sigma^2\{V - X(X'V^{-1}X)^{-1}X'\}.$$

Theorem 12.2

$SSE(\hat\beta_G) = \mathbf{e}_G'V^{-1}\mathbf{e}_G$ satisfies $E\,SSE(\hat\beta_G) = (n-p-1)\sigma^2$, so

$$\tilde\sigma^2 = \frac{(\mathbf{y}-X\hat\beta_G)'V^{-1}(\mathbf{y}-X\hat\beta_G)}{n-p-1}$$

is unbiased for σ^2, and $\widehat{\mathrm{Var}}(\hat\beta_G) = \tilde\sigma^2 (X'V^{-1}X)^{-1}$.

Proof. $SSE(\hat\beta_G)$ is the residual sum of squares of the transformed model, $\|\mathbf{z} - W\hat\beta_G\|^2$, whose expectation is $(n-p-1)\sigma^2$ (Theorem 6.2). Directly: by Theorem 5.4(a),

$$\begin{aligned} E(\mathbf{e}_G' V^{-1}\mathbf{e}_G) &= \sigma^2 \mathrm{tr}[V^{-1}\{V - X(X'V^{-1}X)^{-1}X'\}] \\ &= \sigma^2\{n - \mathrm{tr}((X'V^{-1}X)^{-1}X'V^{-1}X)\} = \sigma^2(n-p-1). \end{aligned}$$

■

By contrast, the OLS estimate $\hat\sigma^2 = \mathbf{y}'(I-H)\mathbf{y}/(n-p-1)$ has

$$E\hat\sigma^2 = \frac{\sigma^2}{n-p-1}[\mathrm{tr}V - \mathrm{tr}\{(X'X)^{-1}X'VX\}],$$

which in general differs from σ^2, so the usual OLS standard errors $\hat\sigma^2(X'X)^{-1}$ are biased in two ways under correlated errors.

Under normality, all the inference of Chapters 6–8 carries over to the transformed model: for instance $\hat\beta_G \sim N(\beta, \sigma^2(X'V^{-1}X)^{-1})$, $(n-p-1)\tilde\sigma^2/\sigma^2 \sim \chi^2(n-p-1)$, and t and F tests use $\tilde\sigma^2$.

Example 12.1 (Averages with unequal precision)

Suppose $y_i = \mu + \epsilon_i$ where y_i is the average of m_i measurements, so $\mathrm{Var}\,\epsilon_i = \sigma^2/m_i$ and $V = \mathrm{diag}(1/m_i)$. Then $X = \mathbf{1}$ and

$$\hat\mu_G = \frac{\sum m_i y_i}{\sum m_i}, \qquad \mathrm{Var}\,\hat\mu_G = \frac{\sigma^2}{\sum m_i},$$

the grand average of all measurements. The OLS estimate $\bar y$ has variance $\frac{\sigma^2}{n^2}\sum \frac{1}{m_i} \geq \frac{\sigma^2}{\sum m_i}$ by the harmonic–arithmetic mean inequality.

Example 12.2 (AR(1) errors)

If $\epsilon_t = \rho\epsilon_{t-1} + u_t$ with $|\rho| < 1$ and u_t i.i.d. $N(0,\sigma_u^2)$, then $\mathrm{Var}\,\epsilon = \frac{\sigma_u^2}{1-\rho^2}V$ with $V_{st} = \rho^{|s-t|}$. The transformation T^{-1} amounts to the familiar quasi-differencing $y_t - \rho y_{t-1}$, $\mathbf{x}_t - \rho\mathbf{x}_{t-1}$ for $t \geq 2$, with the first observation scaled by $\sqrt{1-\rho^2}$. In practice ρ is unknown and is estimated from OLS residuals (feasible GLS).

Unknown V. V is rarely known. It is modelled with few parameters (equal correlation within clusters, AR(1) in time, variance proportional to a power of a regressor) and estimated from the data, typically from OLS residuals; GLS with the estimate plugged in is **feasible GLS**.

12.3 Selected exercises

Motivation. The exercises prove the facts about OLS under correlated errors that Section 12.2 states, work out weighted least squares for a regression through the origin, and verify the AR(1) transformation of Example 12.2.

Exercise 12.1 (OLS under a general covariance)

Let $\mathbf{y} = X\beta + \epsilon$ with $E\epsilon = \mathbf{0}$, $\operatorname{Var}\epsilon = \sigma^2 V$, V known positive definite and X of full column rank $p+1$. Let $\tilde\beta = (X'X)^{-1}X'\mathbf{y}$ and $\hat\sigma^2 = \mathbf{y}'(I-H)\mathbf{y}/(n-p-1)$.
(a) Derive $E\tilde\beta$ and $\operatorname{Var}\tilde\beta$.
(b) Derive $E\hat\sigma^2$.
(c) Show that if $VX = XQ$ for some matrix Q, then $\tilde\beta = \hat\beta_G$.
(d) Let $V = (1-\rho)I_n + \rho J_n$ with $-1/(n-1) < \rho < 1$ (equicorrelated errors), and let X contain the intercept column. Show that OLS and GLS coincide.

Solution.

(a) $E\tilde\beta = (X'X)^{-1}X'X\beta = \beta$ and

$$\operatorname{Var}\tilde\beta = (X'X)^{-1}X'(\sigma^2 V)X(X'X)^{-1} = \sigma^2(X'X)^{-1}X'VX(X'X)^{-1}.$$

(b) By Theorem 5.4(a) with mean $X\beta$, covariance $\sigma^2 V$ and $(I-H)X = 0$,

$$\begin{aligned} E\{\mathbf{y}'(I-H)\mathbf{y}\} &= \sigma^2 \operatorname{tr}\{(I-H)V\} = \sigma^2\big[\operatorname{tr}V - \operatorname{tr}\{X(X'X)^{-1}X'V\}\big] \\ &= \sigma^2\big[\operatorname{tr}V - \operatorname{tr}\{(X'X)^{-1}X'VX\}\big], \end{aligned}$$

using $\operatorname{tr}(AB) = \operatorname{tr}(BA)$. Dividing by $n-p-1$ gives the formula of Section 12.2.

(c) VX has full column rank (V is nonsingular), so XQ does, and the square matrix Q is nonsingular. Then $V^{-1}X = XQ^{-1}$ and $X'V^{-1} = (Q^{-1})'X'$, so

$$\hat\beta_G = \{(Q^{-1})'X'X\}^{-1}(Q^{-1})'X'\mathbf{y} = (X'X)^{-1}Q'(Q^{-1})'X'\mathbf{y} = (X'X)^{-1}X'\mathbf{y} = \tilde\beta.$$

(d) V has eigenvalue $1+(n-1)\rho$ on $\mathbf{1}$ and $1-\rho$ on $\mathbf{1}^\perp$, both positive, so V is positive definite. If the intercept is the first column, $\mathbf{1} = X\mathbf{e}_1$, so $J_nX = \mathbf{1}\mathbf{1}'X = X\mathbf{e}_1\mathbf{1}'X$ and

$$VX = (1-\rho)X + \rho X\mathbf{e}_1\mathbf{1}'X = X\{(1-\rho)I + \rho\,\mathbf{e}_1\mathbf{1}'X\} = XQ.$$

By (c), $\tilde\beta = \hat\beta_G$: with equicorrelated errors and an intercept, OLS is BLUE, although its usual standard errors and $\hat\sigma^2$ are still biased by (a)–(b). ■

Exercise 12.2 (Weighted least squares through the origin)

Let $y_i = \beta x_i + \epsilon_i$, $i = 1,\dots,n$, with $x_i > 0$, uncorrelated errors, $E\epsilon_i = 0$ and $\operatorname{Var}\epsilon_i = \sigma^2 x_i$.
(a) Show that the GLS estimator is $\hat\beta_G = \bar y/\bar x$, with $\operatorname{Var}\hat\beta_G = \sigma^2/(n\bar x)$.
(b) Find the variance of the OLS estimator $\tilde\beta = \sum x_i y_i / \sum x_i^2$.
(c) Verify directly that $\operatorname{Var}\tilde\beta \ge \operatorname{Var}\hat\beta_G$, with equality iff all x_i are equal.

Solution [Supplement].

(a) $V = \operatorname{diag}(x_i)$ and $X = (x_1,\dots,x_n)'$, so $X'V^{-1}X = \sum x_i^2/x_i = \sum x_i$ and $X'V^{-1}\mathbf{y} = \sum y_i$. By Theorem 12.1, $\hat\beta_G = \sum y_i/\sum x_i = \bar y/\bar x$ and $\operatorname{Var}\hat\beta_G = \sigma^2/\sum x_i = \sigma^2/(n\bar x)$.

(b) $\operatorname{Var}\tilde\beta = \sum x_i^2 \operatorname{Var} y_i/(\sum x_i^2)^2 = \sigma^2 \sum x_i^3/(\sum x_i^2)^2$.

(c) We need $(\sum x_i^2)^2 \le \sum x_i \sum x_i^3$. With $a_i = x_i^{1/2}$ and $b_i = x_i^{3/2}$, this is the Cauchy–Schwarz inequality $(\sum a_i b_i)^2 \le \sum a_i^2 \sum b_i^2$, with equality iff $\mathbf{b}$ is proportional to $\mathbf{a}$, i.e. $x_i = b_i/a_i$ is constant. This agrees with the Gauss–Markov argument of Section 12.2. The GLS estimator is the classical *ratio estimator*, natural when the spread of y grows with x. ■

Exercise 12.3 (The AR(1) transformation)

Let $\epsilon_t = \rho\epsilon_{t-1} + u_t$ $(t = 2, \dots, n)$ with $|\rho| < 1$, $E\epsilon_1 = 0$, $\operatorname{Var}\epsilon_1 = \sigma_u^2/(1-\rho^2)$, and let $u_2, \dots, u_n$ have mean 0, variance σ_u^2, and be uncorrelated with each other and with ϵ_1.
(a) Show that $\operatorname{Cov}(\epsilon_s, \epsilon_t) = \sigma_u^2\rho^{|s-t|}/(1-\rho^2)$.
(b) Let $\eta_1 = \sqrt{1-\rho^2}\,\epsilon_1$ and $\eta_t = \epsilon_t - \rho\epsilon_{t-1}$ for $t \geq 2$, i.e. $\eta = L\epsilon$ for a lower bidiagonal matrix L. Show that $\operatorname{Var}\eta = \sigma_u^2 I_n$.
(c) Deduce that OLS applied to $L\mathbf{y}$ and LX is exactly GLS.

Solution [Supplement].

(a) Each ϵ_s is a linear combination of $\epsilon_1, u_2, \dots, u_s$, so u_t is uncorrelated with ϵ_s for $s < t$.

Step 1. (Variances.) By induction, if $\operatorname{Var}\epsilon_{t-1} = \sigma_u^2/(1-\rho^2)$ then $\operatorname{Var}\epsilon_t = \rho^2\operatorname{Var}\epsilon_{t-1} + \sigma_u^2 = \sigma_u^2/(1-\rho^2)$.

Step 2. (Covariances.) For $h \geq 1$,

$$\operatorname{Cov}(\epsilon_{t+h}, \epsilon_t) = \rho\operatorname{Cov}(\epsilon_{t+h-1}, \epsilon_t) + \operatorname{Cov}(u_{t+h}, \epsilon_t) = \rho\operatorname{Cov}(\epsilon_{t+h-1}, \epsilon_t).$$

Iterating down to $h = 0$ gives $\rho^h\operatorname{Var}\epsilon_t = \sigma_u^2\rho^h/(1-\rho^2)$.

(b) For $t \geq 2$, $\eta_t = u_t$, so these have variance σ_u^2 and are mutually uncorrelated. Also $\operatorname{Var}\eta_1 = (1-\rho^2)\operatorname{Var}\epsilon_1 = \sigma_u^2$ and $\operatorname{Cov}(\eta_1, u_t) = \sqrt{1-\rho^2}\operatorname{Cov}(\epsilon_1, u_t) = 0$. Hence $\operatorname{Var}\eta = \sigma_u^2 I_n$.

(c) With $V_{st} = \rho^{|s-t|}$, (a) says $\operatorname{Var}\epsilon = \frac{\sigma_u^2}{1-\rho^2}V$, and (b) says $L\{\frac{\sigma_u^2}{1-\rho^2}V\}L' = \sigma_u^2 I$, i.e. $LVL' = (1-\rho^2)I$. L is nonsingular (triangular with nonzero diagonal), so $V^{-1} = L'L/(1-\rho^2)$. Then

$$\hat{\beta}_G = (X'V^{-1}X)^{-1}X'V^{-1}\mathbf{y} = (X'L'LX)^{-1}X'L'L\mathbf{y},$$

the OLS estimator for the transformed data $L\mathbf{y}$, LX; the constant $1-\rho^2$ cancels. These are the quasi-differences of Example 12.2. ■